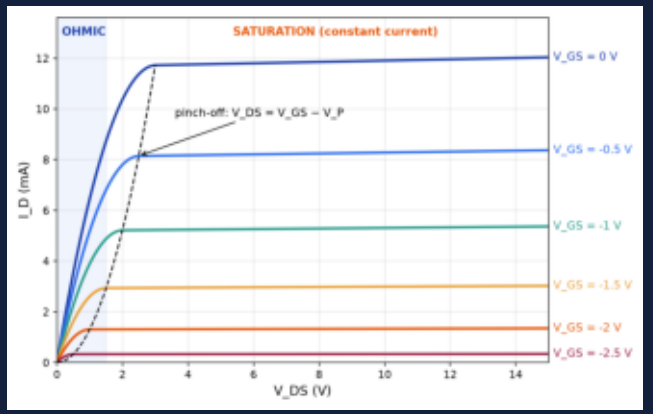
Project · Part B, Q6(b)

Output Characteristics of a JFET

Simulate → Observe → Analyse → Infer

Jayesh Singh · Ayush Kumar · Shikha Saroj · Himanshi Kaushik · Anant Kumar

B.Tech ECE, Section B · KIET Deemed to be University



1 · Problem Statement & Approach

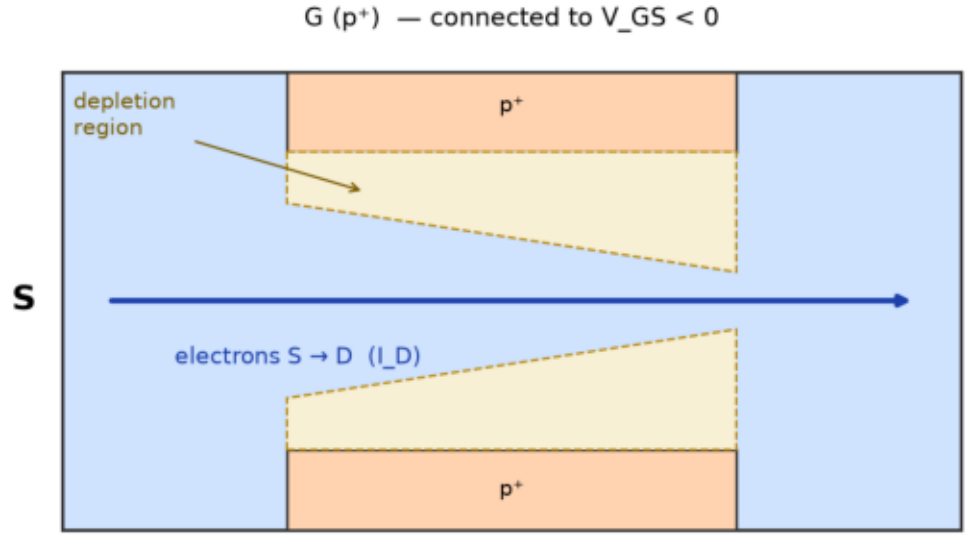
> *“Simulate, observe and analyse the output characteristics of a JFET, infer its behaviour, and explain: (i) JFET is a voltage-controlled current source, (ii) JFET has high input impedance.”*

- **Device:** 2N3819, n-channel JFET (manufacturer SPICE model).
- **Tool:** LTspice (Analog Devices), an industry-standard circuit simulator.
- **Why not Tinkercad?** it has no JFET, and cannot sweep voltages to draw characteristic curves.
- **Plan:** output curves (I_D vs V_DS) → transfer curve (I_D vs V_GS) → gate current I_G.

2 · The JFET: Construction & Working

- **3 terminals:** Source, Drain, Gate. n-channel = n-type bar with p^+ gate regions.
- **Gate junction is reverse-biased** ($V_{GS} \leq 0$), so depletion regions form.
- **More negative V_{GS}** → wider depletion → narrower channel → less I_D.
- **$V_{GS} = V_P$ (≈ -3 V):** channel closes completely → $I_D = 0$ (cut-off).
- **Pinch-off:** depletion is wider at the drain end; at $V_{DS} = V_{GS} - V_P$ the current saturates.

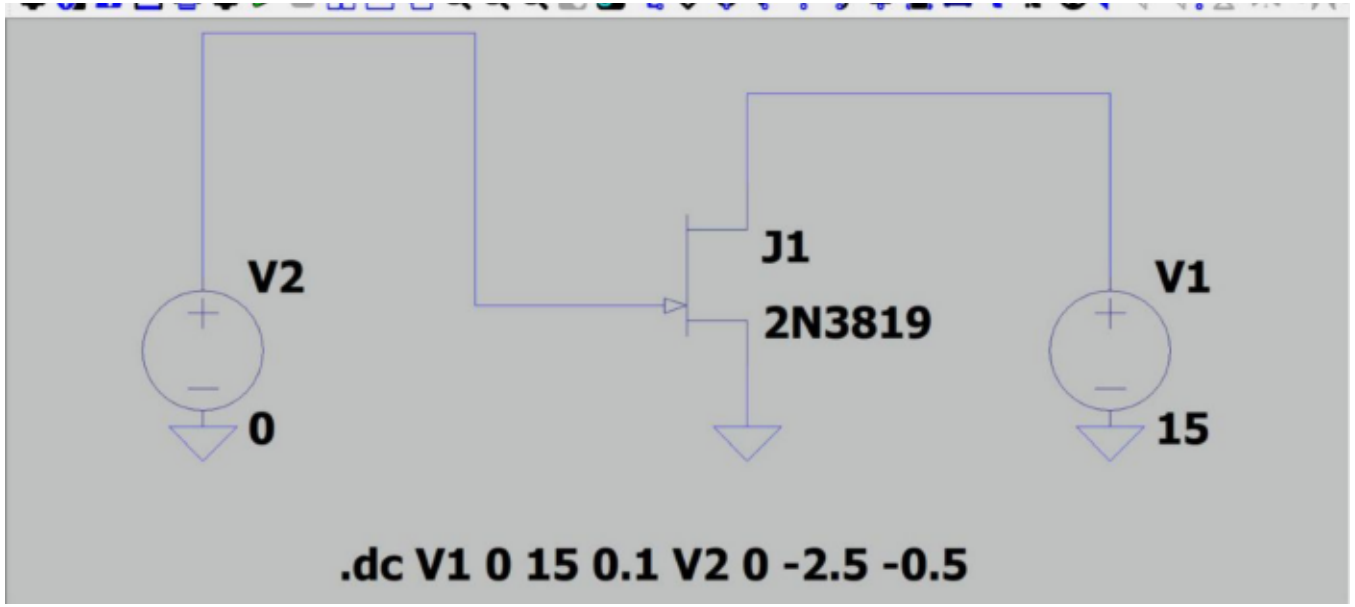
(b) Inside the JFET: gate voltage squeezes the channel



More negative V_GS → wider depletion → narrower channel → less I_D

Depletion is wider at the drain end because V_DS adds to the reverse bias there

3 · Simulation Setup in LTspice



Our LTspice schematic (common-source)

Source	Role
V1	V_DS: swept 0 → 15 V, 0.1 V steps
V2	V_GS: 0, −0.5, −1, −1.5, −2, −2.5 V
J1	2N3819 JFET, source grounded

Command:

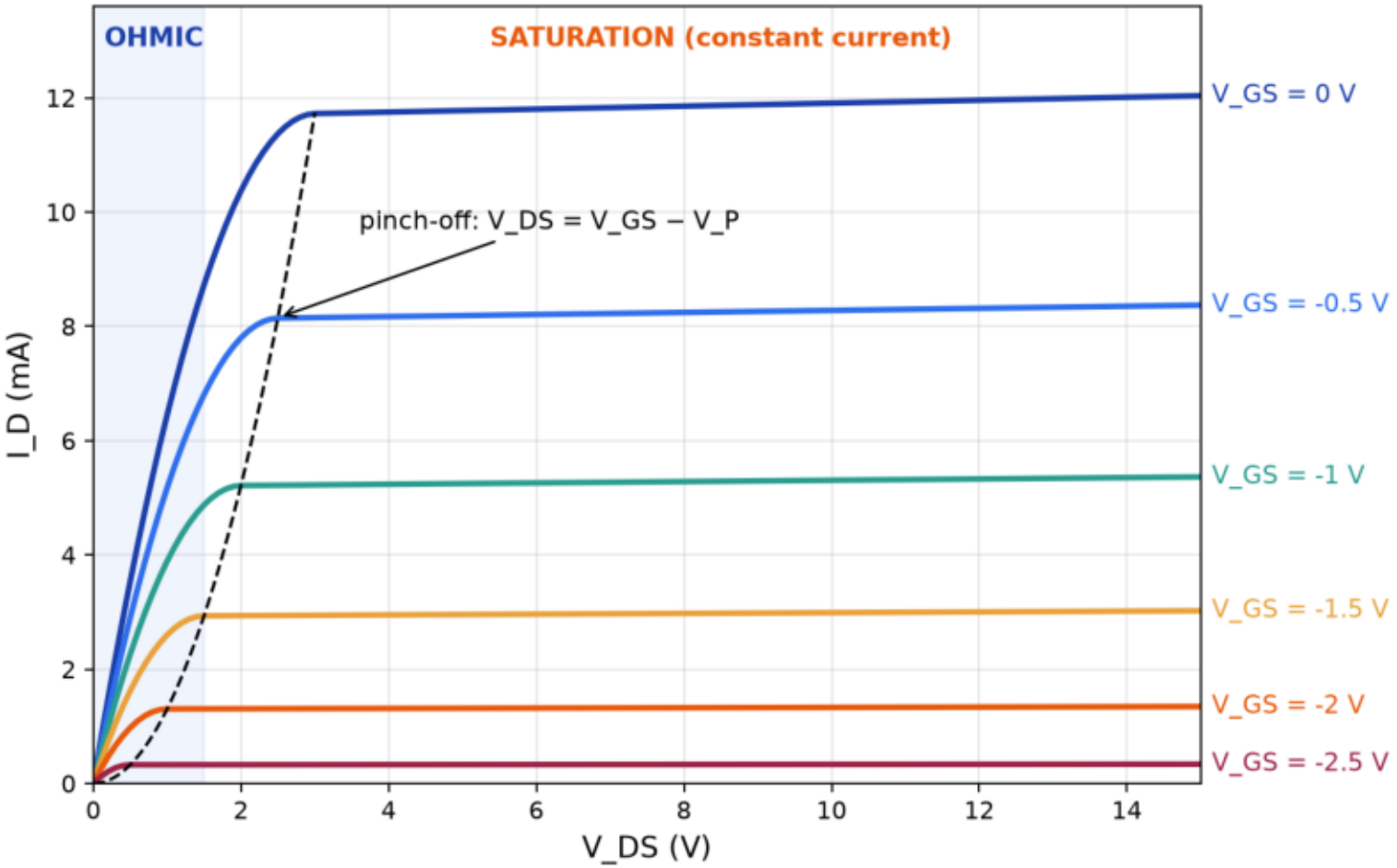
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.dc V1 0 15 0.1 V2 0 -2.5 -0.5
```

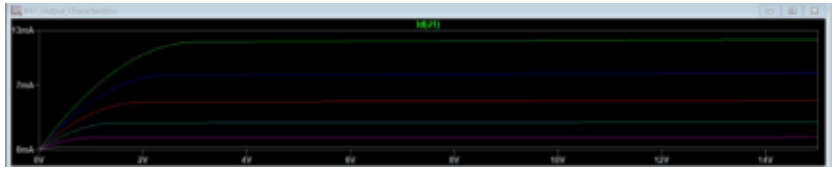
DC sweep of V_DS, repeated for each V_GS:

6 curves × 151 points = 906 operating points.

Second run (transfer curve): V_DS fixed at 10 V, .dc V2 -3.2 0 0.01

4 · Result: Output Characteristics (I_D vs V_DS)





Raw LTspice output

- **Ohmic:** I_D rises with V_DS (voltage-controlled resistor).
- **Pinch-off:** curve bends at V_DS = V_GS − V_P.
- **Saturation:** curves flat; I_D set by V_GS alone.
- **Cut-off:** V_GS ≤ −3 V gives I_D ≈ 0.

5 · Observations from the Simulation Data

V_GS (V)	I_D at V_DS = 5 V	I_D at V_DS = 15 V	Change
0	11.77 mA	12.03 mA	2.2 %
-0.5	8.19 mA	8.37 mA	2.2 %
-1	5.25 mA	5.36 mA	2.2 %
-1.5	2.96 mA	3.02 mA	2.2 %
-2	1.32 mA	1.34 mA	2.2 %
-2.5	0.33 mA	0.34 mA	2.2 %

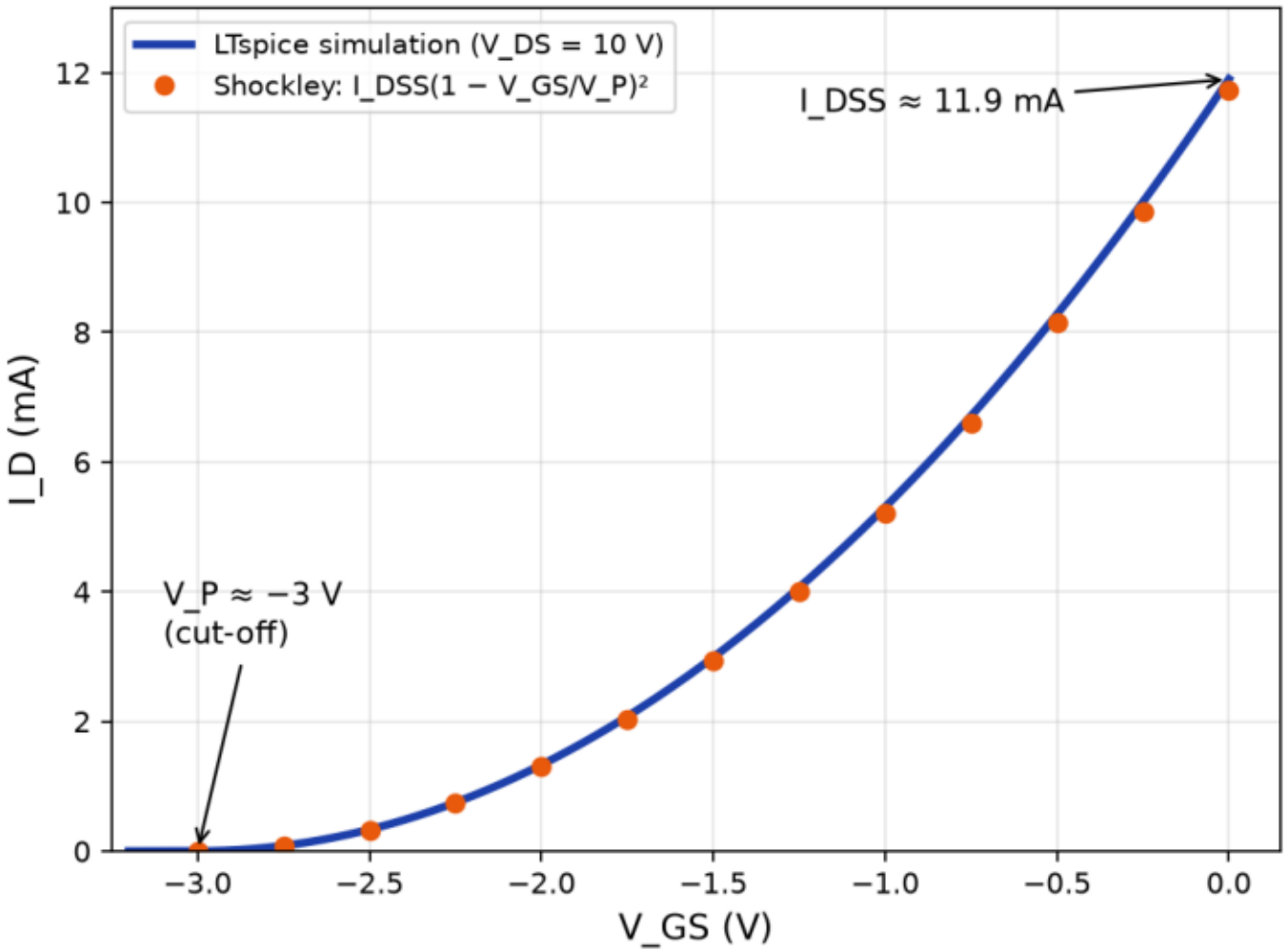
V_DS triples (5 → 15 V) but I_D changes only ~2 %.

V_GS from 0 to −2.5 V cuts I_D 36 times (11.8 → 0.33 mA).

I_DSS (V_GS = 0, saturation) ≈ 11.7–12.0 mA **Pinch-off voltage V_P ≈ −3 V**

Values read directly from the LTspice data export (Id(J1)).

6 · Transfer Characteristic (I_D vs V_GS)



- **Shape:** a parabola, so I_D is set by V_GS.
- **Shockley's equation:** I_D = I_DSS (1 − V_GS / V_P)²
- **Check at V_GS = −1.5 V:** 11.74 × (0.5)² = 2.94 mA; simulation gives 2.99 mA ✔
- **Transconductance:** g_m = ΔI_D/ΔV_GS ≈ 7.8 mS near V_GS = 0 (theory: 2·I_DSS/|V_P| ≈ 7.8 mS).
- **Meaning:** 1 V at the gate controls milliamps at the drain.

7 · (i) JFET is a Voltage-Controlled Current Source

Current source

In saturation the curves are flat: I_D stays almost constant (~2 % change) while V_{DS} goes from 5 V to 15 V. A current source delivers a fixed current regardless of the voltage across it.

…controlled by a voltage

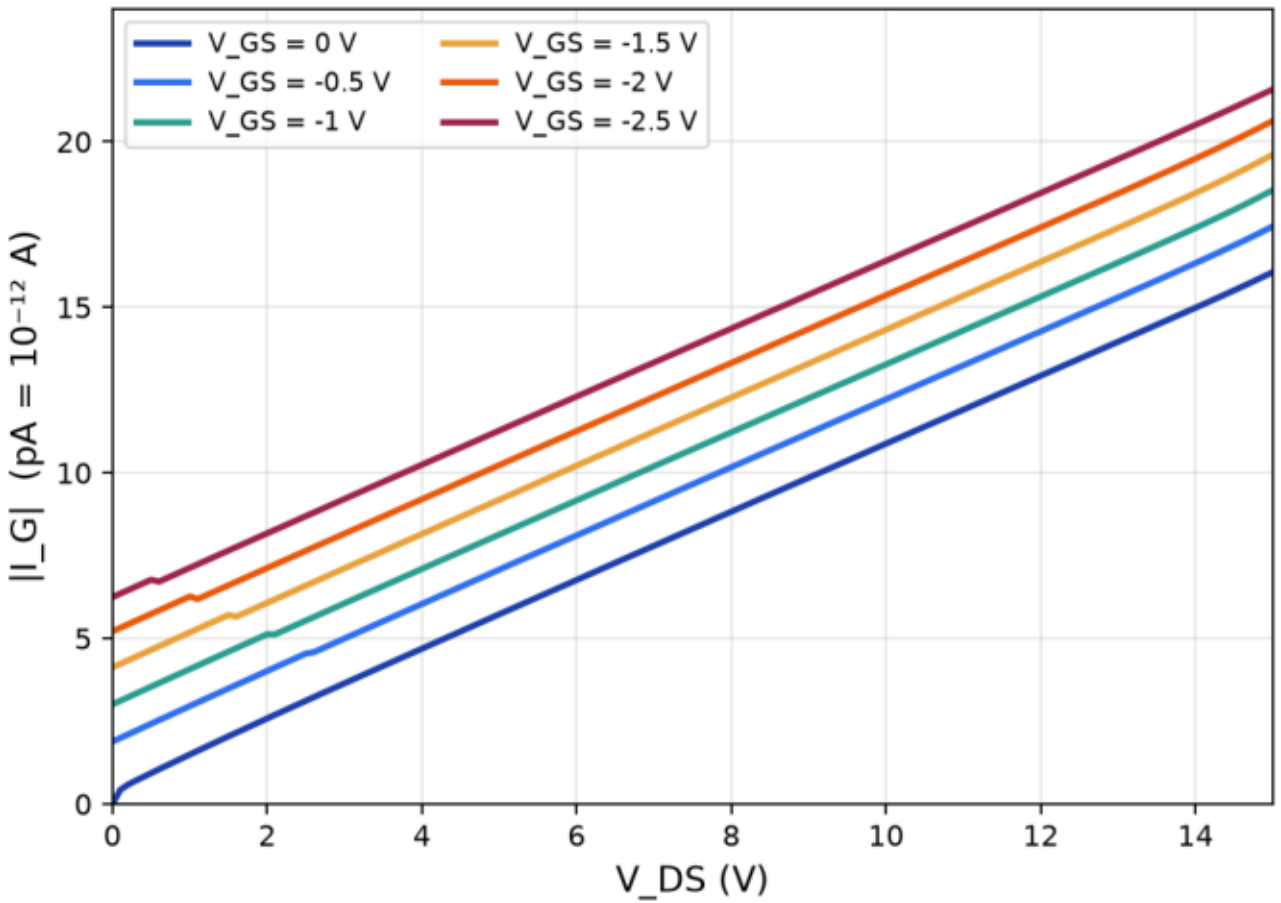
Which flat level you get depends only on V_{GS}: 0 V → 11.8 mA, −1 V → 5.2 mA, −2 V → 1.3 mA. And the gate takes almost no current, so the control is purely by voltage.

$I_D = f(V_{GS})$ and independent of V_{DS} ⇒ VCCS

Small-signal form: $i_d = g_m \cdot v_{gs}$

Contrast: a BJT is current-controlled ($I_C = \beta \cdot I_B$): its input must draw current.

8 · (ii) JFET has High Input Impedance



Simulated gate current I_G (picoamperes)

- **Why:** the gate–channel p–n junction is reverse-biased, so only leakage current flows.
- **Simulation:** largest |I_G| ≈ 21.6 pA (V_GS = −2.5 V, V_DS = 15 V).
- **R_in = V_GS / I_G** = 2.5 V / 21.6 pA ≈ 116 GΩ (~10^{11} Ω).
- **Datasheet:** I_GSS ≤ 2 nA at −15 V → R_in ≥ 7.5 GΩ.
- **Compare:** I_D is mA, I_G is pA: a billion times smaller. A BJT draws µA–mA at its input (kΩ).

So the JFET does not load the signal source.

9 · Conclusions & Applications

- **Output characteristics** show ohmic, saturation and cut-off regions, with $I_{DSS} \approx 11.8$ mA and $V_P \approx -3$ V.
- **Voltage-controlled current source:** in saturation I_D is ~independent of V_{DS} and set only by V_{GS}.
- **High input impedance:** gate current is in picoamperes → $R_{in} \approx 10^{11}\ \Omega$.
- **Behaviour matches theory:** the transfer curve follows Shockley's equation.

Where JFETs are used

Op-amp inputs	Microphones	Test probes	Current sources
TL071 / TL081	inside every electret mic	pH meters, electrometers	constant-current diode

Thank you · Questions?